
Project ARGUS

High-Altitude Platform Station for Hypersonic Cruise Missile Detection

Design Optimization Report | Attritable Swarm Architecture

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Executive Summary

Project ARGUS is an attritable swarm of solar-powered High-Altitude Platform Stations (HAPS) designed to detect and track hypersonic cruise missiles from below. Each node is a lightweight, ultra-low-cost unmanned aircraft operating at 55,000–75,000 ft, carrying commercial off-the-shelf (COTS) infrasound, short-wave infrared (SWIR), and electro-optical sensors. By flying below the threat corridor and looking upward, ARGUS exploits the extreme thermal contrast between a missile's 300–800°C skin and the cold stratospheric background (–57°C).

A multi-objective optimization engine evaluated 46,335 feasible design points across five sensor tiers, three COTS and two military-grade. The recommended configuration—COTS Enhanced—achieves 99% swarm-fused detection probability over 500,000 km² with just 68 aircraft at a total fleet cost of \$507,000, well below the cost of a single cruise missile salvo. This cost asymmetry is the core strategic advantage: adversaries expend more in interceptors than the entire sensor network costs.

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1. Mission & Threat Context

Hypersonic cruise missiles represent one of the most challenging threats in modern air defense. Traveling at Mach 5–10 at altitudes between 65,000 and 100,000 ft, they compress engagement timelines to seconds and fly beneath the coverage floor of many space-based sensors while above the ceiling of most ground-based radar. ARGUS fills this gap by positioning persistent sensor nodes directly in the threat corridor.

Key threat characteristics that drive the ARGUS design:

Thermal signature: Aerodynamic heating at Mach 5+ produces skin temperatures of 300–800°C. Against the stratospheric background at –57°C (216.65 K), this creates an enormous infrared contrast ratio detectable even by inexpensive SWIR sensors.

Shockwave propagation: Hypersonic flight generates strong bow shocks that propagate downward as infrasound. MEMS pressure transducers at ARGUS altitudes can detect these acoustic signatures at ranges of 30–75 km depending on sensor quality.

Altitude band: Cruise missiles at 65,000–100,000 ft are above ARGUS nodes (55,000–75,000 ft), so sensors look upward against cold space—ideal for infrared contrast and infrasound reception.

ARGUS is designed for forward deployment in three primary scenarios: carrier strike group defense, theater missile defense along contested coastlines, and wide-area early-warning networks covering up to 500,000 km².

2. Design Philosophy: Attritable Swarm

The central design doctrine is **cost asymmetry**: each ARGUS node must cost less than the cheapest munition an adversary could use to destroy it. If a surface-to-air missile costs \$100,000–\$2,000,000 and an ARGUS node costs \$7,500, the adversary loses money on every engagement. The swarm degrades gracefully—losing nodes reduces coverage density but never creates a single point of failure.

This philosophy drives every design decision: COTS sensors over mil-spec, mesh radio over satellite links, film-laminate airframes over precision composites, Li-S batteries over custom cells, and automated production at runs of 200+ units. The result is a unit cost under \$10,000—roughly the price of a high-end consumer drone.

Swarm fusion compensates for individual sensor limitations. Even with a single-node detection probability of 65%, two overlapping nodes achieve 87.75% independent probability, boosted to 99% with multi-sensor correlation (infrasound + SWIR + EO cross-cueing).

3. Airframe Configuration

All three COTS tiers converge on an identical airframe: 8.0 m wingspan, 0.30 m chord, yielding a wing area of 2.4 m² and an aspect ratio of 26.7. This is a compact, high-aspect-ratio flying wing reminiscent of a large model sailplane, optimized for the ultra-thin air of the lower stratosphere.

Structural Design

The airframe uses carbon fiber composite construction with film-laminate skins (0.12 kg/m^2 per surface), tubular carbon fiber spars, and foam/balsa ribs at 0.50 m spacing. Total structural mass is 1.92 kg. Spar mass scales with $\text{span}^{2.2}$, reflecting the combined bending and buckling loads. Maximum practical span is capped at 35 m based on structural analysis.

Parameter	Value	Notes
Wingspan	8.0 m	Optimal for COTS payload mass
Chord	0.30 m	Thin airfoil, 12% t/c
Wing area	2.4 m^2	Upper surface: 80% solar coverage
Aspect ratio	26.7	High L/D at low Reynolds number
Structural mass	1.92 kg	CF spar + film-laminate skin
L/D ratio	48.9	Near-theoretical best of 49.3
Cd0	0.008	Ultra-clean, no landing gear drag
Oswald efficiency	0.93	High AR, clean wing
Propulsion	2x BLDC motors	0.9 kg total, 82% prop efficiency

Table 1: ARGUS COTS airframe specifications.

Power System

The upper wing surface carries 1.92 m^2 of GaAs multijunction solar cells (30% efficiency) producing a peak of 762 W and a usable day-average of 399 W after MPPT and DC-DC losses. Night flight is sustained by Li-S batteries (400 Wh/kg specific energy, 80% DoD). The COTS Enhanced configuration requires 1,379 Wh of battery capacity (3.96 kg), with a power margin of 117%—meaning the system generates more than double the energy it consumes over a 24-hour cycle.

Solar endurance is effectively unlimited; the practical mission limit of 30 days is set by maintenance scheduling, not energy. Battery-only emergency range is 1,380 km.

4. Sensor Tier Comparison

The optimizer evaluated five sensor tiers spanning two orders of magnitude in cost. The three COTS tiers all meet the sub-\$10,000 unit cost target while achieving high swarm-fused detection probability.

Tier	Unit Cost	Mass	Det. Range	Single Pd	Swarm Pd	Fleet	Fleet Cost
COTS Minimal	\$3,161	7.9 kg	30 km	35%	83%	415	\$1.31M
COTS Standard	\$4,897	9.3 kg	50 km	52%	99%	150	\$735K
COTS Enhanced	\$7,458	11.5 kg	75 km	65%	99%	68	\$507K
Mil Standard	\$91,475	17.8 kg	120 km	82%	99%	27	\$2.47M
Mil Premium	\$235,688	26.3 kg	200 km	94%	99%	14	\$3.30M

Table 2: Comparison of all five sensor tiers for 500,000 km² coverage.

COTS Enhanced achieves the lowest total fleet cost (\$507K) while maintaining 99% swarm-fused detection probability—cheaper than COTS Standard (\$735K) because its longer detection range (75 km vs. 50 km) means each node covers 3.5x more area, requiring far fewer aircraft.

5. Recommended Design: COTS Enhanced

The COTS Enhanced tier represents the optimal balance of capability and cost. Its sensor suite consists entirely of commercially available components:

Component	Cost	Function
MEMS infrasound x8 (precision)	\$500	Shockwave detection array
SWIR InGaAs sensor array	\$1,500	Thermal tracking (1.0–1.7 μm)
NVIDIA Jetson Orin Nano	\$250	On-board ML classification
EO camera (stabilized)	\$300	Visual target verification
Multi-band radio + GPS/INS	\$400	Comms, navigation, timing

Table 3: COTS Enhanced sensor component manifest.

Total sensor cost is \$5,000, with comms hardware adding \$400 (LoRa mesh radio, 200 km LOS range). Payload mass is 2.8 kg, drawing 21 W continuous and 43 W peak. The Jetson Orin Nano enables on-board inference for target classification, reducing mesh bandwidth requirements and enabling autonomous cueing between infrasound detection and SWIR/EO verification.

Flight Performance

Parameter	Value
Cruise speed	51.9 kts (96 km/h)
Stall speed	43.1 kts (80 km/h)
Stall margin	1.20 ($V_{\text{cruise}} / V_{\text{stall}}$)
Mach number	0.090
Reynolds number	82,547
Max wind tolerance	41.5 kts
Operational altitude	55,000 ft (16.8 km)
Max feasible altitude	58,000 ft (17.7 km)
Solar endurance	30 days (maintenance-limited)
Battery-only range	1,380 km
Climb time to altitude	2.3 hours

Table 4: COTS Enhanced flight performance.

6. Fleet Architecture & Coverage

For a 500,000 km² coverage target (approximately the area of Spain, or the Eastern Mediterranean), the COTS Enhanced fleet requires 43 active nodes with 68 total aircraft (including rotation and 20% spares). Each node covers 17,671 km² (75 km detection radius), with a 1.5x overlap factor ensuring every point in the coverage area is observed by at least 2 nodes.

The mesh network operates via LoRa radio with 200 km line-of-sight range at altitude. Nodes relay detection alerts through the mesh to gateway nodes, which can be connected to theater command via existing communications infrastructure. Satellite uplink is not required at the COTS tier, reducing per-unit cost by approximately \$15,000.

The swarm tolerates graceful degradation: losing 20% of nodes reduces overlap from 2x to ~1.6x, dropping fused detection probability from 99% to approximately 93%—still operationally effective. Complete network neutralization would require destroying all 43 active nodes, costing the adversary roughly \$4.3M in interceptors (at \$100K per SAM) to defeat a \$507K sensor network.

7. Cost Analysis & Asymmetry

Per-Unit Cost Breakdown

Cost Item	Amount	% of Total
Sensor payload	\$5,400	72.4%
Airframe (CF composite)	\$384	5.1%
Assembly, integration, test	\$500	6.7%
Propulsion (2x BLDC)	\$300	4.0%
Power electronics	\$200	2.7%
Recovery hardware	\$200	2.7%
Tooling (amortized/200)	\$190	2.5%
Battery (Li-S)	\$207	2.8%
Solar cells	\$77	1.0%
Total per unit	\$7,458	100%

Table 5: Per-unit cost breakdown for COTS Enhanced.

The sensor payload dominates cost at 72.4% of the total. The airframe itself—the physical aircraft—costs just \$384 in materials at production scale (200 units, automated CF layup at \$200/kg). This confirms that ARGUS is fundamentally a sensor delivery platform; the aircraft is nearly free relative to what it carries.

Cost Asymmetry Analysis

Scenario	ARGUS Cost	Adversary Cost	Ratio
Single node vs. short-range SAM	\$7,458	\$100,000	1 : 13
Single node vs. medium-range SAM	\$7,458	\$500,000	1 : 67
Full fleet vs. full engagement	\$507,111	\$4,300,000+	1 : 8+
Full fleet vs. single cruise missile	\$507,111	\$1,000,000+	1 : 2+

Table 6: Cost asymmetry across engagement scenarios.

Annual maintenance cost is \$50,711 (10% of fleet value). Expected airframe lifetime is 3.0 years, limited by stratospheric UV and thermal cycling degradation of the film-laminate skin. Total 3-year lifecycle cost including spares and maintenance is approximately \$659,000.

8. Feasibility & Constraints

The COTS Enhanced design passes all seven feasibility checks in the optimization engine:

Check	Criterion	Actual	Status
Energy feasible	Day surplus > night consumption	3,789 Wh surplus	PASS
Power margin	> 5%	117%	PASS
Stall margin	$V_{\text{cruise}}/V_{\text{stall}} > 1.15$	1.20	PASS
Wing loading	< 5.0 kg/m ²	4.79 kg/m ²	PASS
Mass reasonable	< 250 kg total	11.5 kg	PASS
Aspect ratio	10 – 40	26.7	PASS
Span structural	≤ 35 m	8.0 m	PASS

Table 7: Feasibility checks for COTS Enhanced at 55,000 ft.

The primary altitude constraint is the exponential growth of power requirements in thinner air. At 55,000 ft the COTS Enhanced design has a 117% power margin; at 58,000 ft this drops to feasible but tighter; above 61,000 ft the mass convergence loop fails as battery mass spirals. The optimizer uses iterative mass convergence with under-relaxation (factor 0.30) to detect and avoid these divergence conditions.

The stall margin of 1.20 is tight but acceptable for the calm stratospheric environment, where turbulence is minimal and wind speeds rarely exceed 15 m/s. The theoretical limit with $CL_{\text{cruise}} = 0.9$ and $CL_{\text{max}} = 1.3$ is $\sqrt{1.3/0.9} = 1.20$, meaning the design operates near its aerodynamic optimum.

Appendix A: Optimization Methodology

This appendix documents the physics models, equations, and numerical methods used to generate the COTS Enhanced design point. All computations are implemented in `argus_engine.py` and validated against published data for the Airbus Zephyr S HAPS.

A.1 Atmosphere Model

The International Standard Atmosphere (ISA) / US Standard Atmosphere 1976 is implemented with seven layers up to 86 km. At the ARGUS operational altitude of 55,000 ft (16,764 m), the atmosphere is in the tropopause layer (isothermal at 216.65 K):

$$\begin{aligned} T &= 216.65 \text{ K} = -56.5 \text{ deg C (constant in tropopause)} \\ P &= P_{\text{base}} * \exp(-g * dh / (R * T)) \\ &= 101325 * \exp(-9.807 * 16764 / (287.058 * 216.65)) \\ &= 9,120 \text{ Pa (9.0\% of sea level)} \\ \rho &= P / (R * T) = 9120 / (287.058 * 216.65) \\ &= 0.1466 \text{ kg/m}^3 \text{ (12.0\% of sea level)} \end{aligned}$$

The extremely low density is the fundamental driver of HAPS design: lift requires either very large wing area, very high speed, or very high CL. ARGUS uses $CL_{\text{cruise}} = 0.9$ (near CL_{max}) to minimize wing area and thus structural mass.

A.2 Aerodynamic Model

Cruise speed from lift equation ($L = W$ in level flight):

$$\begin{aligned} V_{\text{cruise}} &= \sqrt{2 * W / (\rho * S * CL)} \\ &= \sqrt{2 * 11.49 * 9.807 / (0.1466 * 2.4 * 0.9)} \\ &= 26.67 \text{ m/s} = 96.0 \text{ km/h} = 51.9 \text{ kts} \end{aligned}$$

Drag coefficient (parabolic polar):

$$\begin{aligned} CD_{\text{induced}} &= CL^2 / (\pi * AR * e) \\ &= 0.9^2 / (\pi * 26.67 * 0.93) = 0.01040 \\ CD_{\text{total}} &= Cd0 + CD_{\text{induced}} = 0.008 + 0.01040 = 0.01840 \\ L/D &= CL / CD = 0.9 / 0.01840 = 48.9 \end{aligned}$$

Power required for level flight:

$$\begin{aligned} D &= 0.5 * \rho * V^2 * S * CD = 2.303 \text{ N} \\ P_{\text{cruise}} &= D * V / \eta_{\text{prop}} = 2.303 * 26.67 / 0.82 = 74.9 \text{ W} \end{aligned}$$

Propeller efficiency of 0.82 is typical for slow, large-diameter propellers at low Reynolds number. The Cd0 of 0.008 reflects an ultra-clean airframe with no landing gear, no exposed sensors, and laminar flow over much of the wing.

A.3 Structural Model

Structural mass is estimated using semi-empirical scaling laws calibrated against the Airbus Zephyr S (25 m span, approximately 32 kg airframe at 75 kg MTOW):

$$\text{Skin mass} = S * \rho_{\text{skin}} * 2 = 2.4 * 0.12 * 2 = 0.576 \text{ kg}$$

$$\text{Spar mass} = \rho_{\text{spar}} * L_{\text{ref}} * (b/L_{\text{ref}})^{2.2}$$

$$= 0.10 * 25 * (8/25)^{2.2} = 0.204 \text{ kg}$$

$$\text{Rib mass} = n_{\text{ribs}} * m_{\text{rib}} = 17 * 0.020 = 0.340 \text{ kg}$$

$$\text{Wing mass} = 0.576 + 0.204 + 0.340 = 1.120 \text{ kg}$$

$$\text{Fuselage} = \text{wing} * 0.18 = 0.202 \text{ kg}$$

$$\text{Tail} = \text{wing} * 0.10 = 0.112 \text{ kg}$$

$$\text{Systems} = \text{airframe} * 0.06 = 0.086 \text{ kg}$$

$$\text{Landing gear} = 0.40 \text{ kg (belly skid)}$$

$$\text{Total structural} = 1.92 \text{ kg}$$

The spar span exponent of 2.2 (rather than the theoretical 2.0 for pure bending) accounts for buckling resistance requirements that grow faster than bending loads. This was validated against PHASA-35 data (35 m span, ~55 kg structure).

A.4 Power System Model

Solar energy chain:

$$\text{Solar area} = S * 0.80 = 2.4 * 0.80 = 1.92 \text{ m}^2$$

$$\text{Atm transmission at 16.8 km} = 0.95 + 0.04 * (16764/30000) = 0.972$$

$$\text{Peak irradiance} = 1361 * 0.972 = 1323 \text{ W/m}^2$$

$$\text{Avg sun factor} = 0.637 * \cos(35 \text{ deg}) * 1.1 = 0.574$$

$$\text{Avg irradiance} = 1323 * 0.574 = 760 \text{ W/m}^2$$

$$\text{Avg solar power} = 1.92 * 760 * 0.30 = 438 \text{ W}$$

$$\text{Usable (after MPPT * DC-DC)} = 438 * 0.96 * 0.95 = 399 \text{ W}$$

Day/night energy balance:

$$\text{Total consumption} = P_{\text{cruise}} + P_{\text{payload}} = 74.9 + 21.0 = 95.9 \text{ W}$$

$$\text{Day energy generated} = 399 * 12.5 \text{ h} = 4,988 \text{ Wh}$$

$$\text{Night consumption} = 95.9 * 11.5 \text{ h} = 1,103 \text{ Wh}$$

$$\text{Day surplus} = 4,988 - (95.9 * 12.5) = 3,789 \text{ Wh}$$

$$\text{Battery required} = 1,103 / 0.80 \text{ (DoD)} = 1,379 \text{ Wh}$$

$$\text{Battery mass} = (1379 / 400) * 1.15 \text{ (overhead)} = 3.96 \text{ kg}$$

Power margin = (day_gen - 24h_consumption) / 24h_consumption = 117%. This large margin provides resilience against cloud cover, winter solar angles, and sensor peak-power events.

A.5 Iterative Mass Convergence

Aircraft mass and power are coupled: heavier aircraft require more cruise power, which requires more battery, which increases mass. This positive feedback loop is solved iteratively with under-relaxation:

$$m_{\text{psm}}(n+1) = \alpha * m_{\text{computed}} + (1 - \alpha) * m_{\text{psm}}(n)$$

$$\alpha = 0.30 \text{ (under-relaxation factor)}$$

$$\text{Convergence criterion: } |m_{\text{computed}} - m_{\text{guess}}| < 0.01 \text{ kg}$$

The solver runs up to 120 iterations with divergence detection: if computed power system mass exceeds 400 kg or total mass exceeds 600 kg, the design point is flagged as infeasible and abandoned. This prevents the solver from wasting time on physically impossible configurations (typically high altitude + heavy payload combinations).

For the COTS Enhanced design at 55,000 ft, convergence is reached in approximately 15 iterations. The fixed mass (structure + payload + propulsion) is 5.62 kg, and the converged power system mass is 5.87 kg, yielding a total of 11.49 kg.

A.6 Swarm Detection Fusion

Independent detection probability for N overlapping nodes:

$$P_{\text{independent}} = 1 - (1 - P_{\text{single}})^N$$

$$= 1 - (1 - 0.65)^2 = 1 - 0.1225 = 0.8775$$

With multi-sensor correlation boost:

$$P_{\text{fused}} = \min(0.99, P_{\text{independent}} + \text{boost} * (N - 1))$$

$$= \min(0.99, 0.8775 + 0.18 * 1) = \min(0.99, 0.99) = 0.99$$

The fusion boost of 0.18 per additional overlapping node reflects the improvement from cross-cueing: an infrasound detection triggers SWIR slew, which triggers EO verification. This sequential cueing across modalities reduces false alarm rates and increases true positive probability beyond the independent sensor assumption.

A.7 Fleet Sizing

Coverage per node (circular footprint):

$$A_{\text{node}} = \pi * R_{\text{det}}^2 = \pi * 75^2 = 17,671 \text{ km}^2$$

$$\text{Effective coverage} = A_{\text{node}} / \text{overlap} = 17,671 / 1.5 = 11,781 \text{ km}^2$$

$$\text{Active nodes} = \text{ceil}(500,000 / 11,781) = 43$$

$$\text{Total fleet} = \text{ceil}(43 * 1.3 * 1.20) = 68$$

The 1.3 rotation factor accounts for transit time and maintenance windows. The 1.2 spares fraction provides attrition reserve. Fleet cost = $68 * \$7,458 = \$507,111$.

A.8 Cost Model

The cost model distinguishes between COTS and military builds. For COTS (payload cost < \$10,000), automated CF layup at \$200/kg, flexible Si solar at \$40/m², COTS electronics at \$200, and flat \$500 AIT. At production runs of 200 units, tooling (\$30,000 + \$1,000/m wingspan = \$38,000) amortizes to \$190/unit. Li-S battery cost is \$150/kWh at volume.

A.9 Optimization Search Space

The optimizer performs a grid search over four dimensions: wingspan (4–35 m in 2 m steps), chord (0.2–1.5 m in 0.1 m steps), altitude (55,000–75,000 ft in 1,000 ft steps), and sensor tier (5 tiers). This yields approximately 280,000 candidate design points, of which 46,335 pass all feasibility checks. For each feasible design, the solver also runs sensitivity analysis by perturbing wingspan, chord, and altitude around the optimum.

Optimization criteria: designs are ranked first by feasibility (all seven checks must pass), then by per-unit cost (lowest wins) within each sensor tier. The maximum-altitude design for each tier is also reported separately.